

View Only

QUESTIONS

Q1

Default and Parameterized Constructors
20 marks

Q2

Constructor Overloading
20 marks

Q3

Copy Constructor with Dynamic Memory
20 marks

Q4

Destructor and Resource Management
20 marks

Q5

Overloading the + Operator (Complex Numbers)
20 marks

5/5 answered

Q1 Default and Parameterized Constructors

20 marks

Write a C++ program that defines a class Book with attributes title, author, and price. Implement:

- A. A default constructor that initializes all attributes to default values ("Unknown", "Unknown", 0.0).
- B. A parameterized constructor that accepts values for all three attributes.
- C. A display() member function to print the book details.

In main(), create one object using each constructor and display its details.

Your Code

```
1 #include <iostream>
2 #include <string>
3 using namespace std;
4
5 class Book
6 {
7     string title;
8     string author;
9     float price;
10
11 public:
12
13
14     Book()
15     {
16         title = "Unknown";
17         author = "Unknown";
18         price = 0.0;
```

Input

stdin input

Output

Visible Test Cases

Test Case 1

Test Case 2

```

19     },
20
21
22     Book(string t, string a, float p)
23     {
24         title = t;
25         author = a;
26         price = p;
27     }
28
29
30     void display()
31     {
32         cout << "Title: " << title << endl;
33         cout << "Author: " << author << endl;
34         cout << "Price: " << price << endl;
35     }
36 };
37
38 int main()
39 {
40
41     Book b1;
42
43
44     Book b2("C++ Programming", "Bjarne Stroustrup", 500);
45
46     cout << "Default Constructor:" << endl;
47     b1.display();
48
49     cout << endl;
50
51     cout << "Parameterized Constructor:" << endl;
52     b2.display();

```


CO3_AT2_Medium Level Class Test

View Only

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5/5 answered

Q2 Constructor Overloading

20 marks

Write a C++ program that defines a class Rectangle with the following constructors:

- A. A default constructor that sets length and breadth to 1.
- B. A single-parameter constructor that creates a square (length = breadth).
- C. A two-parameter constructor that sets both length and breadth.

Include a member function area() to compute the rectangle's area. In main(), create three objects using each constructor and print their areas.

Your Code

```
1 #include <iostream>
2 using namespace std;
3
4 class Rectangle
5 {
6     int length, breadth;
7
8 public:
9     Rectangle()
10    {
11        length = 1;
12        breadth = 1;
13    }
14     Rectangle(int side)
15    {
16        length = side;
17    }
18 }
```

Input

stdin input

Output



Visible Test Cases

```
15     {
16         length = side;
17         breadth = side;
18     }
19
20
21     Rectangle(int l, int b)
22     {
23         length = l;
24         breadth = b;
25     }
26
27
28     int area()
29     {
30         return length * breadth;
31     }
32 };
33
34 int main()
35 {
36     Rectangle r1;
37     Rectangle r2(5);
38     Rectangle r3(6, 4);
39
40     cout << "Area of default rectangle: " << r1.area() << endl;
41     cout << "Area of square: " << r2.area() << endl;
42     cout << "Area of rectangle: " << r3.area() << endl;
43
44     return 0;
45 }
```

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5/5 answered

Q3 Copy Constructor with Dynamic Memory

20 marks

Write a C++ program that defines a class Student containing a dynamically allocated character array for the student's name and an integer for marks. Implement:

- A. A parameterized constructor that allocates memory and copies the name.
- B. A deep copy constructor that allocates new memory for the copied object.
- C. A destructor that releases the allocated memory.
- D. A display() function to print the student's details.

In main(), create a Student object, create another object using the copy constructor, and modify the second object's name to demonstrate that the original remains unaffected.

Your Code

```
1 #include <iostream>
2 #include <cstring>
3 using namespace std;
4
5 class Student
6 {
7     char *name;
8     int marks;
9
10 public:
11     Student(const char n[], int m)
12     {
13         name = new char[strlen(n) + 1];
14         strcpy(name, n);
15         marks = m;
16     }
17 }
```

Input

Jaya 95

Output



Visible Test Cases

Test Case 1


```

18 Student(const Student &s)
19 {
20     name = new char[strlen(s.name) + 1];
21     strcpy(name, s.name);
22     marks = s.marks;
23 }
24
25 void setName(const char n[])
26 {
27     delete[] name;
28     name = new char[strlen(n) + 1];
29     strcpy(name, n);
30 }
31
32 void display()
33 {
34     cout << "Name: " << name << endl;
35     cout << "Marks: " << marks << endl;
36 }
37
38 ~Student()
39 {
40     delete[] name;
41 }
42 };
43
44 int main()
45 {
46     char name[50];
47     int marks;
48
49     cin >> name >> marks;
50
51     Student s1(name, marks);

```



```
43  
44 int main()  
45 {  
46     char name[50];  
47     int marks;  
48  
49     cin >> name >> marks;  
50  
51     Student s1(name, marks);  
52     Student s2(s1);  
53  
54     cout << "Original Student:" << endl;  
55     s1.display();  
56  
57     cout << "Copied Student:" << endl;  
58     s2.display();  
59  
60     s2.setName("Modified");  
61  
62     cout << "After modifying copied object:" << endl;  
63     cout << "Original Student:" << endl;  
64     s1.display();  
65  
66     cout << "Copied Student:" << endl;  
67     s2.display();  
68  
69     return 0;  
70 }
```


Management
20 marks

Q5

Overloading the + Operator
(Complex Numbers)
20 marks

5/5 answered

```
5 {  
6     string name;  
7  
8     public:  
9         FileHandler(string n)  
10        {  
11            name = n;  
12            cout << "File " << name << " opened." << endl;  
13        }  
14  
15        ~FileHandler()  
16        {  
17            cout << "File " << name << " closed." << endl;  
18        }  
19    };  
20  
21    int main()  
22    {  
23        {  
24            FileHandler f1("File1");  
25            FileHandler f2("File2");  
26            FileHandler f3("File3");  
27        }  
28  
29        cout << "All files are closed." << endl;  
30  
31        return 0;  
32    }
```

Output



Visible Test Cases

Test Case 1

Test Case 2

Test Case 3


```
18     temp.real = real + c.real;
19     temp.imag = imag + c.imag;
20     return temp;
21 }
22
23 void display()
24 {
25     cout << real << " + " << imag << "i" << endl;
26 }
27 };
28
29 int main()
30 {
31     int r1, i1, r2, i2;
32
33     cin >> r1 >> i1;
34     cin >> r2 >> i2;
35
36     Complex c1(r1, i1);
37     Complex c2(r2, i2);
38
39     Complex c3 = c1 + c2;
40
41     cout << "Result: ";
42     c3.display();
43
44     return 0;
45 }
```

Test Case 2

Test Case 3